

CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 13: Statistics

Every formula and result of the chapter on one sheet

1 Class mark

$$x_i = \frac{\text{lower limit} + \text{upper limit}}{2}$$

2 Class size

$$h = \text{upper limit} - \text{lower limit}$$

3 Mean: direct method

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

4 Mean: assumed mean method

$$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}, \text{ where } d_i = x_i - a$$

5 Mean: step-deviation method

$$\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i} \right) \times h, \text{ where } u_i = \frac{x_i - a}{h}$$

6 Mode

$$l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

7 Median

$$l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$$

8 Modal class

The class with the highest frequency

9 Median class

The class whose cumulative frequency is just greater than $\frac{n}{2}$

10 Empirical relationship

$$3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$$

11 Making classes continuous

Subtract 0.5 from each lower limit and add 0.5 to each upper limit (for a gap of 1)